Homework - Question Set 3

1. A number X is selected from $\{1, 2, \ldots, 2n-1\}$. Find $E(X)$ and $Var(X)$.

2. Let X have the pdf

$$f(x) = \begin{cases} 2(1-x) & 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

   Find $E(X)$, $E(X^2)$ and $E(6x + 3x^2)$.

3. Let us divide, at random, a horizontal line segment of length 5 into 2 parts. Let X be length of left-hand part with pdf

$$f(x) = \begin{cases} 1/5 & 0 < x < 5 \\ 0 & \text{elsewhere.} \end{cases}$$

   Find $E(X)$, $E(5-X)$ and $E(X(5-X))$.

4. Let X have the pdf

$$f(x) = \begin{cases} \frac{1}{2}(x+1) & -1 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

   Find mean and variance.

5. Suppose we have a fair spinner with the numbers 1, 2, and 3 on it. Let X be the number of spins until the first 3 occurs. Assuming that the spins are independent, the pmf of X is

$$p(x) = \frac{1}{3}\left(\frac{2}{3}\right)^{x-1}, \quad x = 1, 2, 3, \ldots$$

   Find the moment generating function. Using mgf find mean and variance.